Conjugate-pair solves: solutions at infinity

The 6 missing solutions are 3-point decompositions at infinity; a random projective chart recovers every 4-point decomposition at the same cost

2026-11-06. This follows `lift_dense_conjugate_pair_decomp_20261106`. There, on curve 1, the counted k = 4 Gaudry solver found quotient dimension 4,090 instead of 4,096 on both conjugate-pair targets that decompose, and returned nothing. This study asks two questions:

1. Why are 6 of the 4,096 solutions lost?
2. Does a solver that does not assume 4,096 affine solutions recover the decompositions?

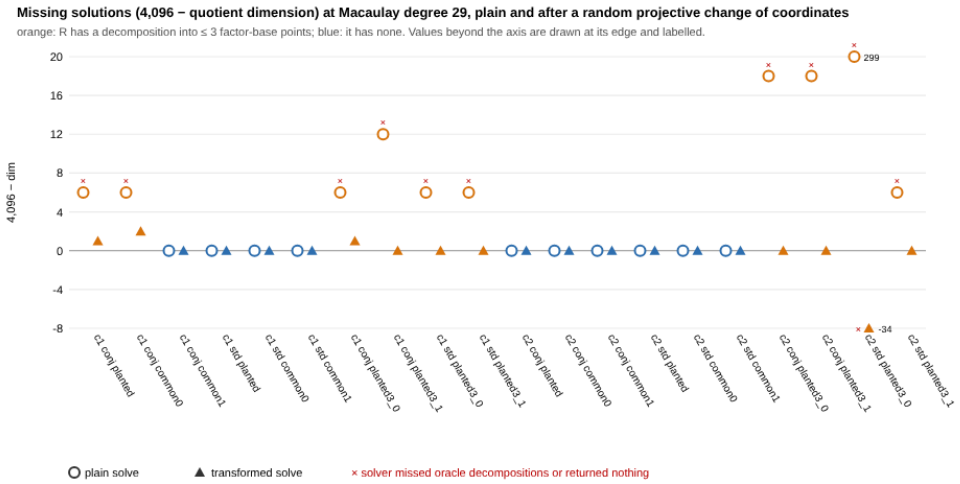
- **Protocol.** Frozen before any data at `44edbfbc20`. Amendment A1 was registered before its own data at `0b64bace85` (`PROTOCOL.md`).
- **Instrument.** `examples/conjpair_decomp_cost.rs` `--inf`, at `f5dcb899a4`. It adds a random projective change of coordinates, a short-decomposition oracle and 3-point-planted targets. `gaudry_quartic` is called unchanged.
- **Provenance.** The run was resumed after the 2-hour task limit. The commits for each part are in `reports/run-20261106T1550Z/commit.txt`.
- **Host.** This container, 4 cores and 15 GB. Iso Lab EKS needs an AWS identity this container does not have. The metric is a deterministic count, and the A/A rule passes.
- **Report.** `REPORT.pdf`.

Every statement below is an unverified assertion of this session [pcc:unverified]. The claim record is at the end.

Method

- **Setup.** p = 29, the same 2 curves, lines and H as the previous study. The system is 4 Weil components of degree 8 in e = (e_1, …, e_4).
- **Transform.** Homogenise the system with e_0 and substitute (e_0, …, e_4) = A·(y_0, …, y_4), with A ∈ GL_5(F_p) drawn uniformly from a seeded stream. Then dehomogenise at y_0 = 1 and solve with the unchanged `solve_system4_with` (chains, sparsest, Macaulay degree 29).
- **Mapping back.** Each rational solution maps back by (e_0, e) = A·(1, y). If $e_0 \neq 0$ it is an affine solution e/e_0, which is checked against the original components. If $e_0 = 0$ it is a point at infinity.
- **Oracles.** An exact meet-in-the-middle gives the 4-point decompositions. A second oracle counts the 1-, 2- and 3-point multisets of the line's factor base that sum to R.
- **Runs** (40 solves at degree 29, plus 6 for A1) [pcc:unverified]:
  - **R1:** the previous study's 12 targets, plain.
  - **R2:** the same 12 targets, transformed.
  - **R3:** 8 new targets, each planted as a sum of 3 factor-base points (2 per arm and curve), solved both plain and transformed.

Answer to question 1: the missing solutions are shorter decompositions, sent to infinity



4,096 − dim per target, plain and transformed

- **The deficit appears exactly when a short decomposition exists** (rule Q1-a) [pcc:unverified].
  - On all 20 plain solves, dim < 4,096 if and only if R is a sum of at most 3 points of the line's factor base.
  - There are 0 exceptions. Both of the previous study's failing targets are among the 20; each has exactly one 3-point decomposition.
  - Every target with no short decomposition has dim 4,096, and the solver returns its decompositions.
- **The mechanism.**
  - The leading coefficient of S_5 in x_1 is $S_4(x_2, x_3, x_4, x_R)^2$.
  - So a decomposition $R = \pm P_2 \pm P_3 \pm P_4$ is a solution of the Weil system with one coordinate sent to O, which means one e-coordinate at infinity.
  - Those points of the Bézout count leave the affine chart. The plain Macaulay solve then reports "border unreachable" (or, on curve 2, a broken chain structure) and returns nothing, even though some 4-point decompositions are still affine.
- **This is generic, not specific to conjugate-pair lines** (rule Q1-c, PASS).
  - Targets planted as 3-point sums lose dimension on the ordinary F_p line as well (dim 4,090), and the plain solver fails there in the same way.
  - Curve 1's conj line hit this on random targets because a conjugate-pair base carries many more relations, so short ones too. The previous study measured about 36× more relations per trial.
- **The size of the deficit** (observed, no rule).
  - Eight targets have exactly one 3-point decomposition and no shorter one. The deficit is 6 on six of them, 12 on curve 1 conj `planted3_1`, and 18 on curve 2 conj `planted3_0`.
  - It is 18 on curve 2 conj `planted3_1`, which has three.
  - On curve 2 std `planted3_0` it is 299. That is the one target that is itself a factor-base point (short1 = 1); see below.

Answer to question 2: the transformed solve recovers every 4-point decomposition

- **19 of 20 transformed solves succeed** [pcc:unverified].
  - On all 19, the affine solutions equal the oracle's 4-point decompositions exactly (rule Q2).
  - That includes the two curve 1 targets that failed before, with 3/3 and 2/2 found. It also includes the 3-point-planted targets with 4-point decompositions (2/2, 1/1, 2/2 and 3/3).
- **The 20th fails** on curve 2 std `planted3_0[0, 14, 14]`.
  - That R is itself a factor-base point, so R = R + T + (−T) for every T on the line. That is a one-dimensional family of decompositions, a positive-dimensional component of the solution set.
  - Plain, the component lies at infinity (dim 3,797). After the transform it enters the affine chart: dim 4,130 > 4,096 at degree 29, and the solver reports "border unreachable".
  - That target has no 4-point decomposition, so Q2 passes there only vacuously.
  - This explanation is derived after the data and was not registered.
- **The 3-point decompositions appear at infinity** (rule Q1-d, report only).
  - Every successful transformed solve with a short decomposition returns at least one rational point with $e_0 = 0$.
  - Where short3 = 3 there are 3 such points. Some targets have more points at infinity than 3-point multisets (2 against 1 on curve 1 conj `planted3_1`, 3 against 1 on curve 2 conj `planted3_0`).
- **Cost: the same as a generic solve** (rule Q2-cost, PASS).

- The 12 R2 solves cost 0.970–1.030× the main run's mean generic plain solve on the same curve and arm. That is $1.124–1.191 \cdot 10^{11}$ F_p multiplications.
- The transform itself costs $2.3–2.9 \cdot 10^{6}$ multiplications, about $2 \cdot 10^{-5}$ of a solve.

curve	arm	target	plain dim	plain solver/oracle	transformed dim	transformed solver/oracle	transformed / generic cost
1	conj	planted	4,090	**0/3**	4,095	3/3	1.028
1	conj	common 0	4,090	**0/2**	4,094	2/2	1.016
1	conj	common 1	4,096	0/0	4,096	0/0	0.970
1	std	planted	4,096	1/1	4,096	1/1	1.030
2	conj	planted	4,096	3/3	4,096	3/3	1.012
1	conj	planted3_0	4,090	**0/2**	4,095	2/2	—
1	std	planted3_0	4,090	**0/1**	4,096	1/1	—
2	conj	planted3_0	4,078	**0/2**	4,096	2/2	—
2	conj	planted3_1	4,078	**0/3**	4,096	3/3	—
2	std	planted3_0	3,797	0/0	**4,130**	0/0, failed	—

This table shows a selection. All 40 rows, with outcomes and short counts, are in `SUMMARY.md`.

Amendment A1: the remaining shortfall is not truncation

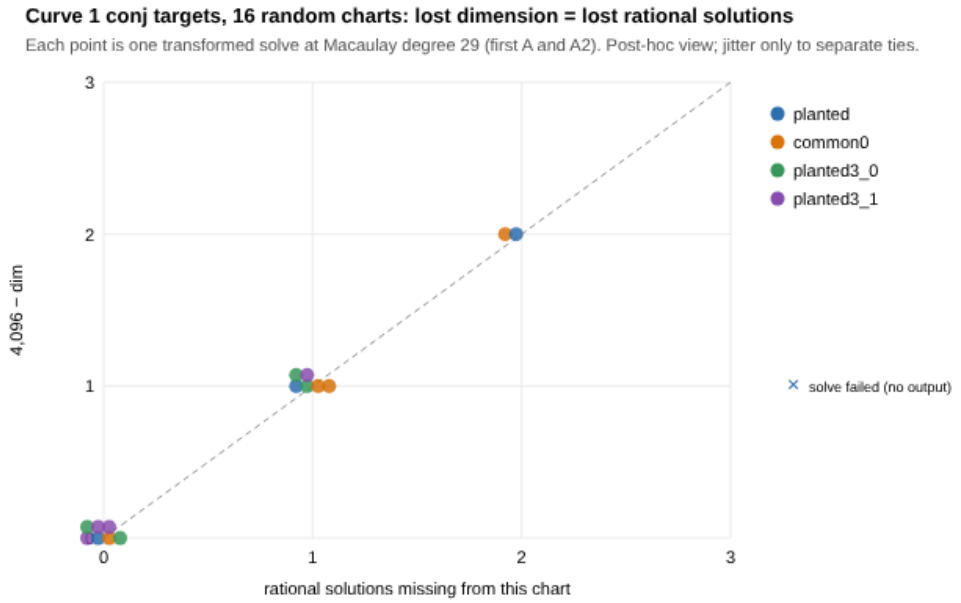
On curve 1's two degenerate conj targets, the transformed system has dim 4,095 and 4,094, not 4,096, even though the solver returns every decomposition. Rule M2 asked whether a higher Macaulay degree closes that gap [pcc:unverified]:

target	degree	matrix	dim	F_p muls	solver/oracle
planted	30	48,580 × 46,376	4,095	$1.588 \cdot 10^{11}$	3/3
planted	31	56,124 × 52,360	4,095	$2.089 \cdot 10^{11}$	3/3
planted	32	64,458 × 58,905	4,095	$2.712 \cdot 10^{11}$	3/3
common 0	30	48,580 × 46,376	4,094	$1.571 \cdot 10^{11}$	2/2
common 0	31	56,124 × 52,360	4,094	$2.068 \cdot 10^{11}$	2/2
common 0	32	64,458 × 58,905	4,094	$2.685 \cdot 10^{11}$	2/2

- **M2: not truncation.** The dimension is unchanged from degree 29 to 32, so 1 and 2 solutions, counted with multiplicity, are still outside the chart.
- **Candidates, reported but not decided:**
  - a solution that also lies on the new hyperplane at infinity. A random F_p-rational hyperplane contains a given rational point with probability about 1/p;
  - a solution of multiplicity above 1 at infinity;
  - a non-zero-dimensional component.
- **The cheapest test** is a second independent A on the same targets. Amendment A2 below runs it.
- **Decompositions are unaffected.** Every oracle decomposition is recovered at every degree.
- **Q1-b.** The verdict stays as registered at degree 29.

Amendment A2: the remaining shortfall belongs to the chart, not the system

Amendment A2 was registered at `0542602115` before its data. It re-solves each of curve 1's four conj targets with a short decomposition at degree 29, under 3 fresh random charts A_1, A_2 and A_3, each from its own seed. That is 12 solves, about 42 min [pcc:unverified].



lost dimension against lost rational solutions, 16 charts

target	first A	A_1	A_2	A_3
planted	4,095	**4,095, solve failed (0/3)**	4,096	4,094
common 0	4,094	4,096	4,095	4,095
planted3_0	4,095	4,096	4,096	4,095
planted3_1 (control)	4,096	4,096	**4,095**	4,096

- **M3: chart-dependent on every target** [pcc:unverified].
  - Each of the three targets that fell short with the first A reaches 4,096 under at least one fresh chart.
  - So no multiple point or positive-dimensional component is needed. The Bézout count of 4,096 is there, and a given chart can leave part of it on its own hyperplane at infinity.
- **The control fails.** `planted3_1`, at 4,096 with the first A, drops to 4,095 under A_2. That fits the same reading: no chart is safe in advance.
- **The Q2 re-check fails once.**
  - On `planted` under A_1, the transformed solve reports “border unreachable” and returns nothing (0/3), as the plain solve did.
  - The other 11 solves return exactly the oracle’s decompositions. That includes every chart with dim 4,095 or 4,094.
- **What is lost is rational solutions, one for one** (post-hoc, not registered).
  - Within a target, every unit of missing dimension matches one F_p-rational e-solution that the solver no longer returns:
    - `planted`: 16, 15 and 14 rational solutions at dim 4,096, 4,095 and 4,094;
    - `common0`: 15, 14 and 13;
    - `planted3_0` and `planted3_1`: 13 and 12.
  - That is what an F_p-rational solution on the chart’s hyperplane at infinity produces. Each rational solution lies on a random F_p-rational hyperplane with probability about 1/p.
  - With 13–16 rational solutions at p = 29, the chance that a chart misses at least one is about $1 - (28/29)^{16} \approx 0.43$. On these four targets 9 of 16 charts fell short.
  - Every other target’s first chart reached 4,096, apart from the positive-dimensional case. Most of those targets have 0–4 rational solutions. The three curve 2 conj targets with 12–18 had a combined chance of about 0.22 of all avoiding a shortfall, so curve 2 was partly lucky too.
  - The missing rational solutions were never one of the oracle’s decompositions, but there is no reason that should always hold.
- **Q1-b.** The verdict stays as registered.

Registered verdicts (`--summarize-inf`)

rule	verdict	detail
A/A: R1 reproduces the main run’s total_muls and dim	PASS	12/12
Q1-a: plain dim < 4,096 ⇔ a ≤ 3-point decomposition exists	**PASS**	20 plain solves, 0 exceptions
Q1-b: transformed dim = 4,096	**FAIL**	4,095, 4,094 and 4,095 on curve 1 conj (planted, common 0, planted3_0); 4,130 on curve 2 std planted3_0 (positive-dimensional, see above)

rule	verdict	detail
Q1-c: R3 plain dim < 4,096 on both arms	**PASS**	8/8, conj and std
Q2: transformed solver = oracle	**PASS**	20/20; on curve 2 std planted3_0 only vacuously (solve failed, oracle 0)
Q2-cost: R2 within [0.9, 1.1] × generic	**PASS**	0.970–1.030
Q1-d: points at infinity vs short counts (report)	—	at least one per short decomposition on every successful solve
A1 / M2: dim 4,096 at some degree ≤ 32	not reached	4,095 and 4,094 through degree 32: not truncation
A2 / M3: some fresh chart gives 4,096	**chart-dependent** on all 3 targets	4,096 under 1, 1 and 2 of 3 fresh charts
A2 control: planted3_1 stays at 4,096	**FAIL**	4,095 under A_2
A2 Q2 re-check: solver = oracle	**FAIL**	11/12; `planted` under A_1 returned nothing (border unreachable)

What this changes

- **The previous study's curve 1 failure is not a property of conjugate-pair lines.** It is the standard degeneracy of summation-polynomial systems when R has a shorter decomposition. With a random projective chart, the per-trial cost and correctness on conjugate-pair lines match the F_p line on both curves.
- **What this means for the family's relation phase.** The cost study's 11–17× fewer trials, at equal cost per trial, now holds on both curves measured [pcc:unverified]. The previous caveat ("only on curves where the solver handles the system") is removed for targets that are not themselves factor-base points.
- **One random chart is not always enough** (A2). A chart with an F_p-rational solution on its hyperplane at infinity loses that solution. In 1 of 16 charts on curve 1 it also lost every decomposition. The cheap guard is the Bézout check: if dim < 4,096, draw a new A and solve again. On these targets that is about one extra solve in two. On targets with few rational solutions it is rare. A chart defined over F_{p^2} would avoid every F_p-rational point except those on a codimension-2 subspace. It is not tested here.
- **A further gain left unused.** A transformed solve also returns the 3-point relations, at infinity. They are not counted here.
- **Scope.** Two curves at p = 29, one A per target (four on curve 1's conj targets with a short decomposition). No ECDLP cost against rho is claimed, and there is no leaderboard or benchmark row.

Reproduce

```
        cargo build --release --example conjpair_decomp_cost
        B=./target/release/examples/conjpair_decomp_cost
        $B --p 29 --curves 2 --inf --out rows.jsonl                          # ~2.3 h
(resumable: --skip-existing rows.jsonl)
        $B --p 29 --curves 2 --inf --macdeg 30,31,32 --only-arm conj --only-curve 1
\
           --only-decomposable --out a1_rows.jsonl                           # ~56
min
        $B --summarize-inf rows.jsonl a1_rows.jsonl \
           research/lift_dense_conjugate_pair_decomp_20261106/reports/run-
20261106T0643Z/rows.jsonl out/ claimed.json
        python3 research/lift_dense_conjugate_pair_infinity_20261106/fig.py out/   #
layout only
        $B --p 29 --curves 2 --a2 3 --only-curve 1 --only-arm conj --out
a2_rows.jsonl   # ~42 min
        $B --summarize-a2 a2_rows.jsonl rows.jsonl out/ claimed_a2.json
        python3 research/lift_dense_conjugate_pair_infinity_20261106/fig2.py out/  #
layout only
```

Claim record

claim	standing	content
`PCC-20261106-e1a9d3`	not covered (unsigned)	rule counts, cost band, A1 degrees (`validator-report.json`, values in `claimed.json`)
`PCC-20261106-213a67`	not covered (unsigned)	Amendment A2: chart dependence, control, Q2 re-check (`validator-`

claim	standing	content
		report-a2.json, values in claimed_a2.json)